

Speech and Audio Processing

03 - speech synthesis

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Class schedule (tentative)

01	Oct. 03	Introduction	Learn from basic to state-of-the-art about speech processing.
02	Oct. 10	Speech production / 音声生成	
03	Oct. 17	Speech synthesis / 音声合成	
04	Oct. 24	Speech recognition / 音声認識	
05	Oct. 31	Self-supervised learning / 自己教師あり学習	
06	Nov. 7	Foundation model / 基盤モデル	
07	Nov. 14	Sound recognition/synthesis / 環境音認識合成	
08	Nov. 28	Advanced topics / 発展的課題	
09	Dec. 05	Advanced topics / 発展的課題	
10	Dec. 12	Speech system building / 音システム構築	Learn how to build your speech system.
11	Dec. 19	Speech system building / 音システム構築	
12	Dec. 26	Speech system building / 音システム構築	
13	Jan. 09	Speech system building / 音システム構築	
14	Jan. 23	Final presentation / 最終発表	

About this class

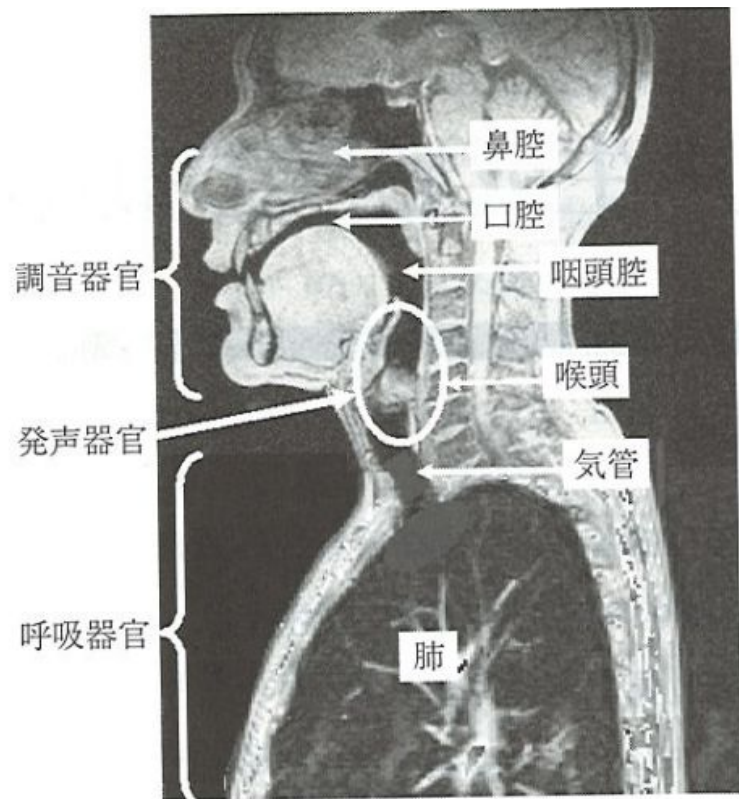
- Language
 - Slides in English (w/ partial translation to Japanese, e.g., named entity)
 - Talk in Japanese (w/ simultaneous translation to English.)
 - Once translation finished for current slide, I will move onto next slide.
 - Either En or Ja is fine for your question.
- Grades
 - 50 / 100 ... final presentation
 - 50 / 100 ... final report
- Room
 - 14-202

Final presentation & reports / 最終発表&レポート

- Perform a unique project that either
 - improves on state-of-the-art in speech-related {signal processing, machine learning} tasks,
 - applies {machine learning, signal processing} to a unique task,
 - investigates human behaviors in human–speech-AI interaction.
 - prototypes the future of human and speech AI.
- Presentation & report
 - Slides in English.
 - Talk in Japanese or English. (choose one you prefer.)
 - I will drive {Ja, En}-to-{En, Ja} simultaneous translation during presentation.
 - Reports in Japanese or English. (choose one you prefer.)

Previous class

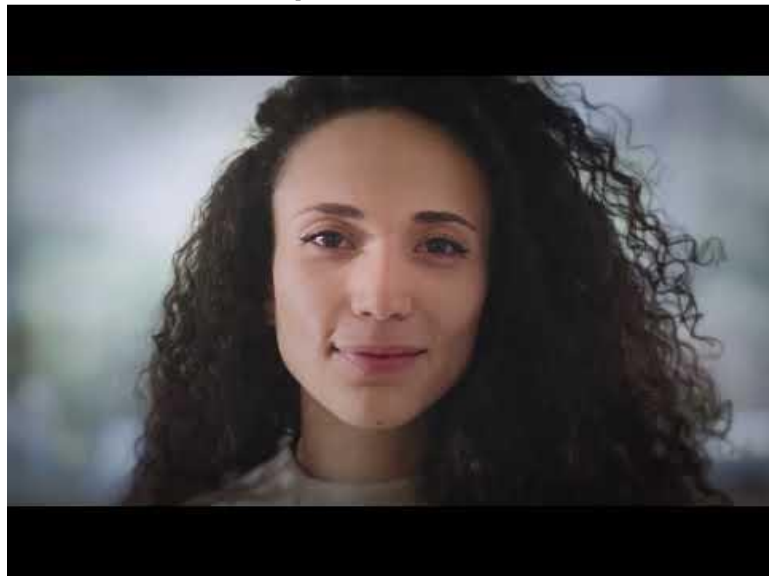
What happens in **our body** when **producing** speech



Today's goal

What happens in **computers** when **artificially synthesizing** speech

Text-to-speech / 音声合成



Singing voice synthesis / 歌声合成



History / 歷史

Voder, the first synthesizer (1939)

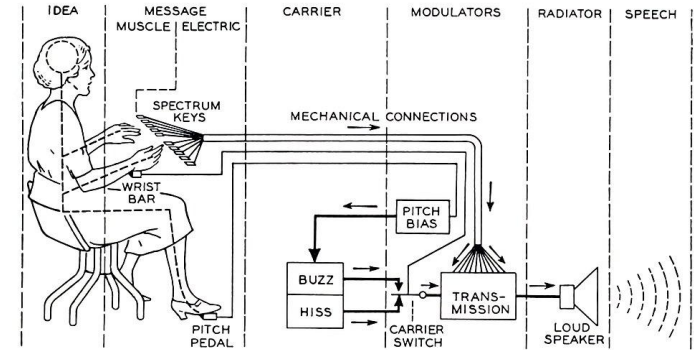


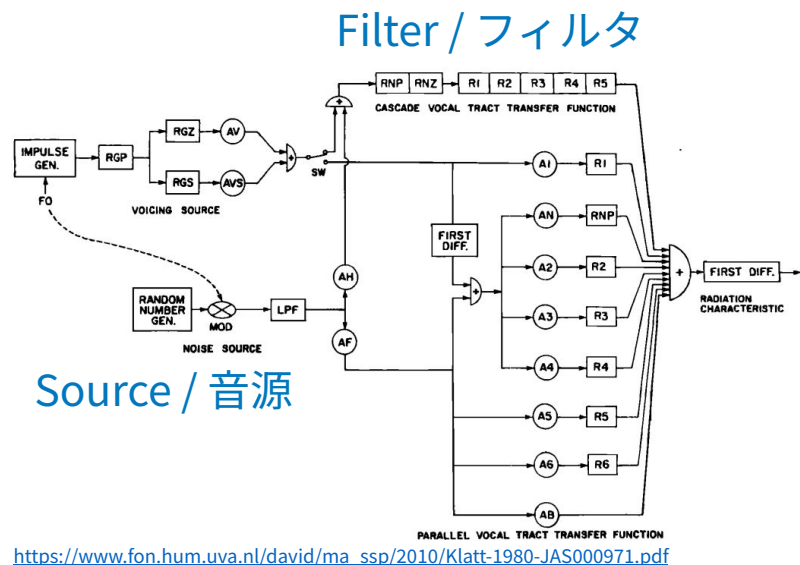
Fig. 8—Schematic circuit of the voder.

<https://en.wikipedia.org/wiki/Voder>

Daisy Bell by world's first “singing computer” (1961)



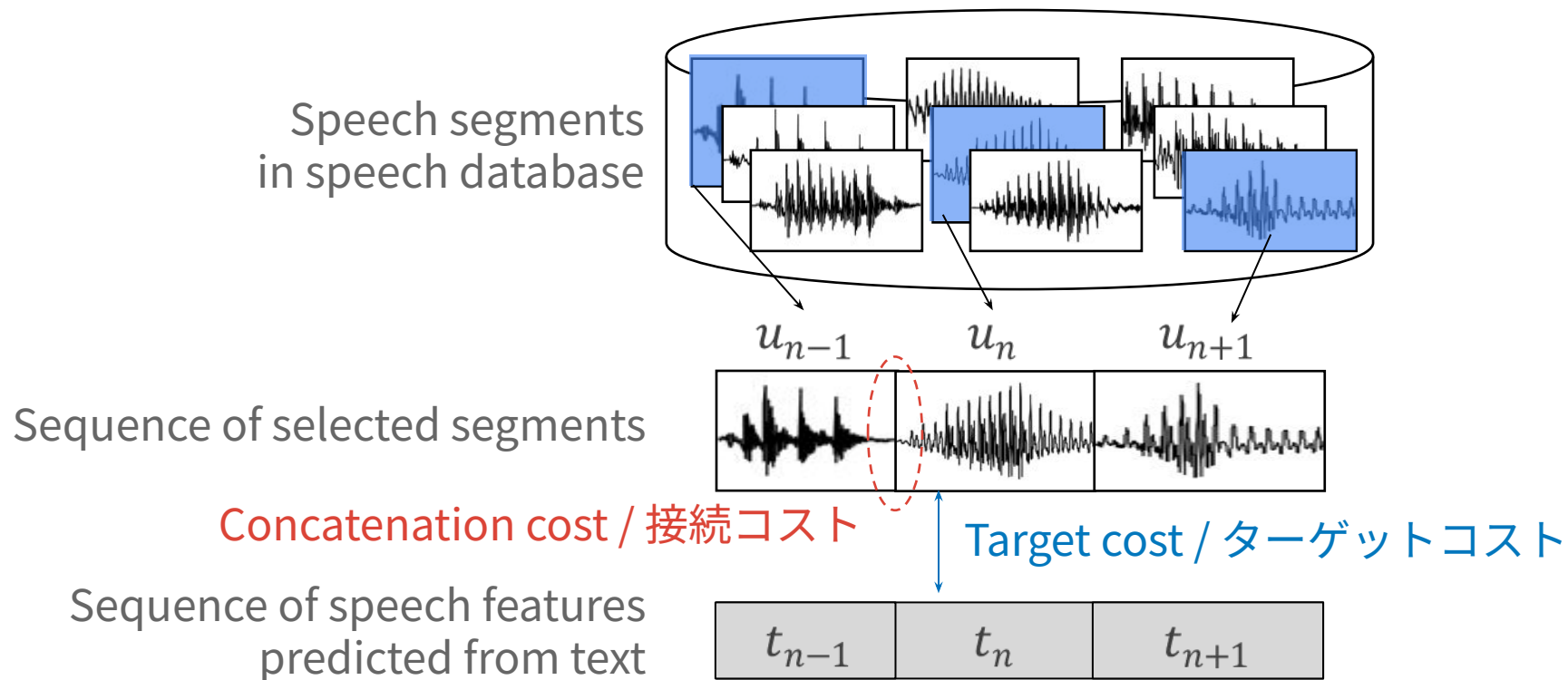
Formant synthesis & DECtalk (1983)



(Formant synthesis still exists in Japanese)



Era of data-driven synthesis: unit selection (素片選択) (1990s)



Selecting segments to minimize total costs including two costs.

VOCALOID: unit selection-based singing voice synthesis



INTERSPEECH 2007



VOCALOID – Commercial singing synthesizer based on sample concatenation

Hideki Kenmochi, Hayato Ohshita

Center for Advanced Sound Technologies, Yamaha Corporation, Japan

https://www.isca-archive.org/interspeech_2007/kenmochi07_interspeech.html



剣持秀紀

@kenmochi フォローされています

歌声合成とかやっています。
嫌いなこと: 「べし」を連体形で終えること。残響が消
「ラデッキー行進曲」→「ラデツキー行進曲」だよ。
好きなもの: フラット5つの調
Bluesky: @kenmochi.com/
mixi: Chirper II/
mixi2: kenmochi

<https://x.com/kenmochi>

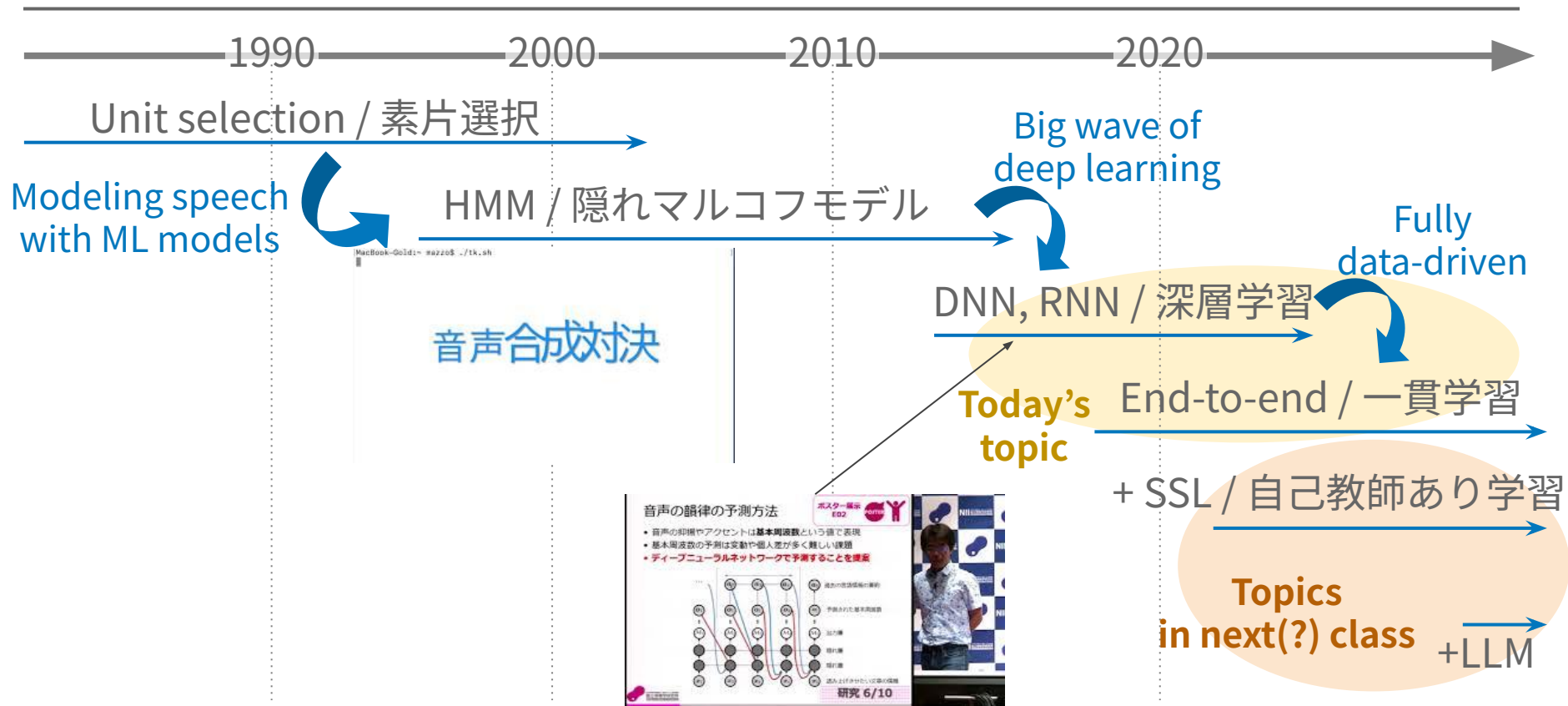


剣持秀紀 @kenmochi · 10月13日
今年はネギを育てています。
私がネギ育てるなんて、なんか面白くないですか？(笑)

<https://x.com/kenmochi/status/1977508491633037347>

Father of VOCALOID

Era of machine learning approaches



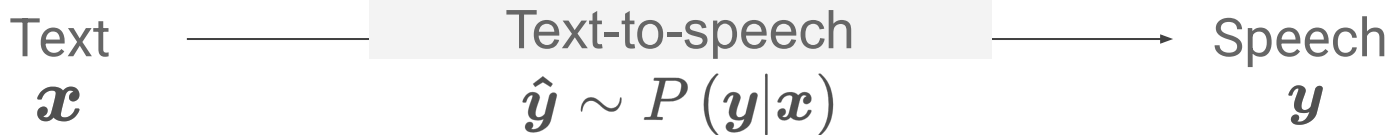
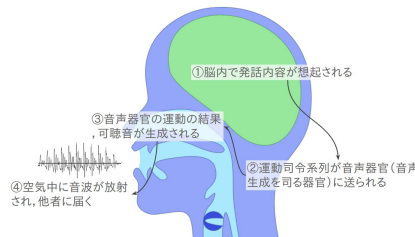
* DTW: dynamic time warping, HMM: hidden Markov model, DNN: deep neural network, RNN: recurrent neural network
VQ: vector quantization, GMM: Gaussian mixture model, SSL: self-supervised learning, LLM: large language model

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Formulation of speech synthesis / 音声合成の定式化

Formulation / 定式化

- A method to artificially synthesize speech.
 - Emulating humans' speech production by computers
- Described as generative models (生成モデル)



Why generative model (probabilistic model)? / なぜ生成モデル (統計モデル) を使うのか？

- Humans' speech production is probabilistic, not deterministic.
/ 人間の音声生成がそもそも確率的であり、決定論的でない
- Motor command / 運動指令:
 - The command always changes even if you pronounce the same word.
- Source & articulation / 音源 & 調音:
 - Source signal is not complete periodic.
- Radiation / 放射:
 - Frequency phase varies when you move.

Mitigating into subproblems / 部分問題への緩和

$P(\mathbf{y}|\mathbf{x})$ Original formulation corresponding to “end-to-end” modeling.

$$= \int \int P(\mathbf{y}|\mathbf{o}, \mathbf{l}, \mathbf{x}) \underbrace{P(\mathbf{o}, \mathbf{l}|\mathbf{x})}_{\text{Introducing intermediate features: speech feats } \mathbf{o}, \text{ and linguistic feats } \mathbf{l}.} d\mathbf{o} d\mathbf{l}$$

$$= \int \int P(\mathbf{y}|\mathbf{o}, \mathbf{l}, \mathbf{x}) \underbrace{P(\mathbf{o}|\mathbf{l}, \mathbf{x}) P(\mathbf{l}|\mathbf{x})}_{\text{Approximating with point estimates (点推定)}} d\mathbf{o} d\mathbf{l}$$

$$\approx \underbrace{P(\mathbf{y}|\mathbf{o}, \mathbf{l}, \mathbf{x})}_{\text{Assuming statistical independence of } \mathbf{y} \text{ on } \{\mathbf{l}, \mathbf{x}\}} \underbrace{P(\mathbf{o}|\mathbf{l}, \mathbf{x}) P(\mathbf{l}|\mathbf{x})}_{\text{Assuming statistical independence of } \mathbf{o} \text{ on } \mathbf{l}.}$$

$$= P(\mathbf{y}|\mathbf{o}) P(\mathbf{o}|\mathbf{l}) P(\mathbf{l}|\mathbf{x})$$

Mitigating into subproblems / 部分問題への緩和

$$P(\mathbf{y}|\mathbf{o})P(\mathbf{o}|\mathbf{l})P(\mathbf{l}|\mathbf{x})$$

Prob. of waveform
given by speech feats.
(vocoder / ボコーダ)

Prob. of speech feats.
given by linguistic feats.
(acoustic model
/ 音響モデル)

Prob. of linguistic feats.
given by text
(text analysis / テキスト解析)

Handling waveform-level
fluctuation, e.g.,
aperiodicity (非周期性)

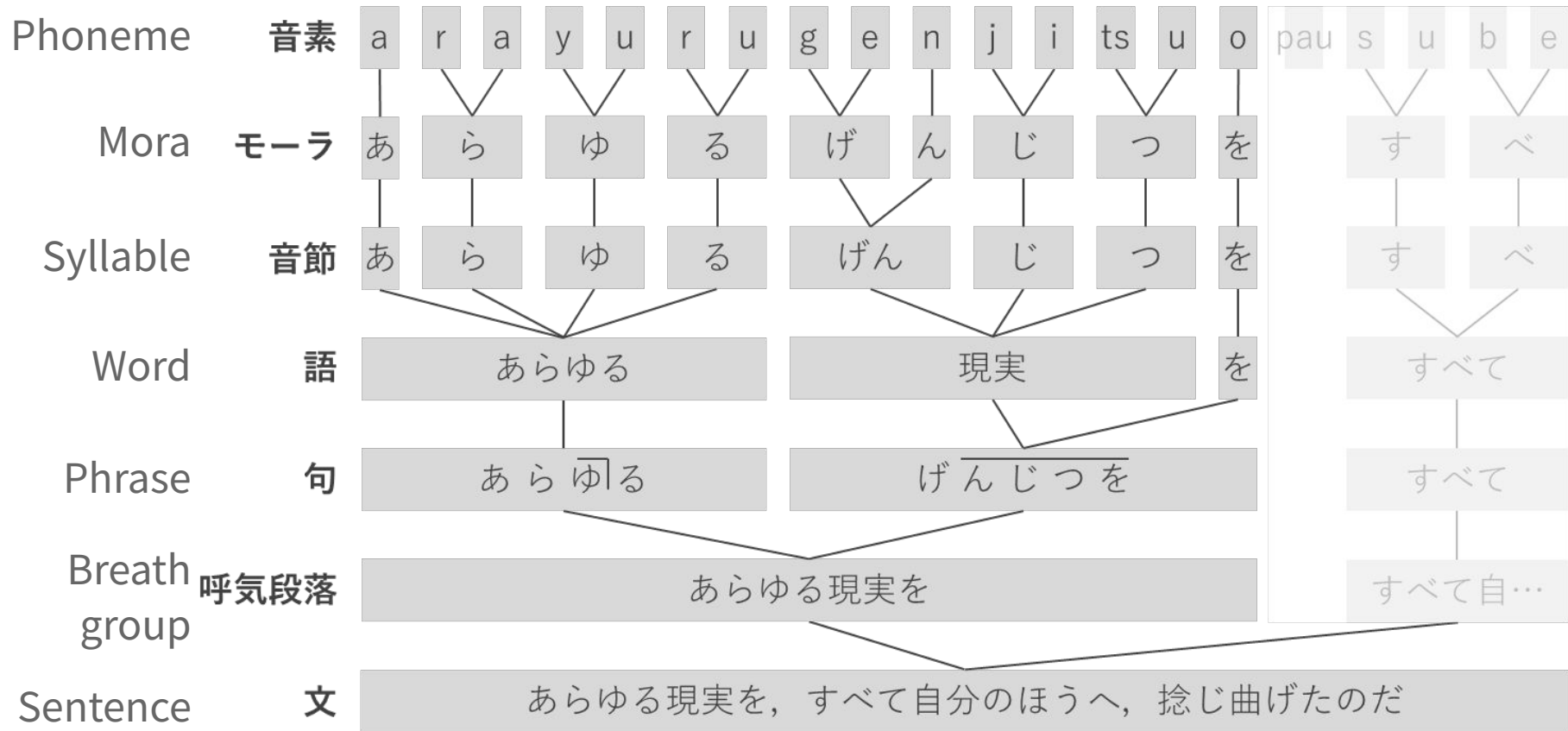
Handling command-level
or feat.-level fluctuation,
e.g., movement variation.

Handling linguistic feat.-level
fluctuation,
e.g., accent, pronunciation.

Text analysis in Japanese / 日本語のテキスト解析

Linguistic units in Japanese

/ 日本語における言語単位



Phoneme / 音素

- Phoneme / 音素
 - Smallest units that make one word different from another.
/ 意味の違いをもたらす, 音韻的な最小単位
- Vowel (母音, V) /a, i, u, e, o/
 - Phonologically: The nucleus of a syllable that can form a syllable alone or in combination with consonants. / 音韻的には「単独あるいは子音を伴って音節を構成する」主音
- Consonant (子音, C) /p, b, t, d, k, g, c, s, z, h, m, n, r, w, j, N, q/
 - Phonemes other than vowels / 母音以外の音

Mora (sub-syllable) and syllable / モーラと音節

- Mora / 拍
 - Smallest unit of timing. / タイミングにおける最小単位
 - Different from syllables, “ー” (長音符), “っ” (促音), “ん” (撥音) are counted as a unique mora
- Syllable / 音節
 - Basic unit defined as a vowel with optional consonants. / 母音を中心とした、対象言語の音声の塊。

表 2-1 日本語における音節とモーラの違い

語 Word	音節 Syllable	モーラ Mora
おはよう o-ha-yo-u	お／は／よ／う	お／は／よ／う
かんぬき ka-n-nu-ki	かん／ぬ／き	か／ん／ぬ／き
いっぱつ i-p-pa-tsu	いっ／ぱ／つ	い／っ／ぱ／つ

Isochrony / 等時性

Any languages can be divided into equal rhythmic positions.

/ 全ての言語は、等しい時間間隔で現れる言語単位がある。

Japanese: mora-timed language / モーラ等時性言語

わたしはとしょかんへいきました。



Mandarin: syllable-timed language / シラブル等時性言語

我 去 图 书 馆



English: stress-timed language / ストレス等時性言語

I went to the library to study for the exam.



Word and accent / 語, アクセント

- Word / 語
 - Linguistic unit consisting of 1+ morphemes (形態素)
- Accent / アクセント
 - Relative emphasis of a certain syllable (or mora) / ある音節 (or モーラ) が卓立すること
 - **Japanese (Tokyo dialect) is a pitch-accent language**
 - Two level (**L**ow or **H**igh) accent.
 - Pitch-fall positions changes the meaning of words. / ピッチの下がる箇所です語の意味が変わる
 - N-mora words have N+1 kinds of accents.
 - type 0 (no pitch-fall): LHH(H)
 - type 1: HLL(L)
 - type 2: HHL(L)

Accent table / アクセント表

“\” indicates pitch-fall positions. “-” indicates no pitch-fall positions.

表 2-3 日本語のアクセント。“\”がアクセント核(下がる位置)、“-”はアクセント核のないことを表します。

	1拍語 1 mora	2拍語 2 morae	3拍語 3 morae	4拍語 4 morae
0型 Type 0: no falling	え- (柄) pattern	はし- (端) edge	さくら- (桜)	ともだち- (友達)
1型 Type 1: fall at 1st	え\ (絵) painting	は\し (箸) chopstick	か\ぶと (兜)	お\んせい (音声)
2型 Type 2: fall at 2nd	-	はし\ (橋) bridge	ここ\ろ (心)	いろ\がみ (色紙)
3型	-	-	おとこ\ (男)	みずう\み (湖)
4型	-	-	-	いもうと\ (妹)

edge

chopstick

bridge

Excepting type 1,
pitch starts with L.

は し (が)
L
ha shi (ga)

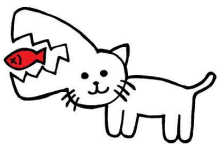
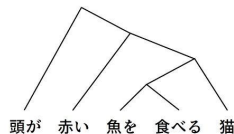
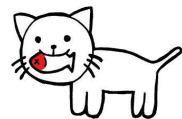
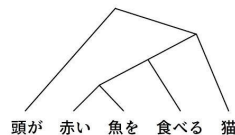
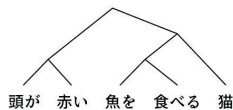
は し (が)

は し (が)

Accent ≠ Intonation

/ アクセントとイントネーションは違う

- Whereas **accent** is determined by words, **intonation** is controlled by intention of the speaker. / **アクセント**は語によって決まるが、**イントネーション**は意図によって制御される。



These examples have the same text, but we can distinguish them by controlling prosody (-> intonation!) / 左の例は全て同じテキストだが、我々は音声で言い分けることが出来る

頭が 赤い 魚を 食べる 猫
atamaga akai sakanawo taberu neko

(off-topic) creole language / クレオール言語

Contact language that develops from the process of different languages.



Gi-lan creole (宜蘭クレオール) by language contact of Japanese and Atayal languages

(off-topic) NHK accent dictionary / NHK アクセント辞典

Japan Broadcasting Corporation



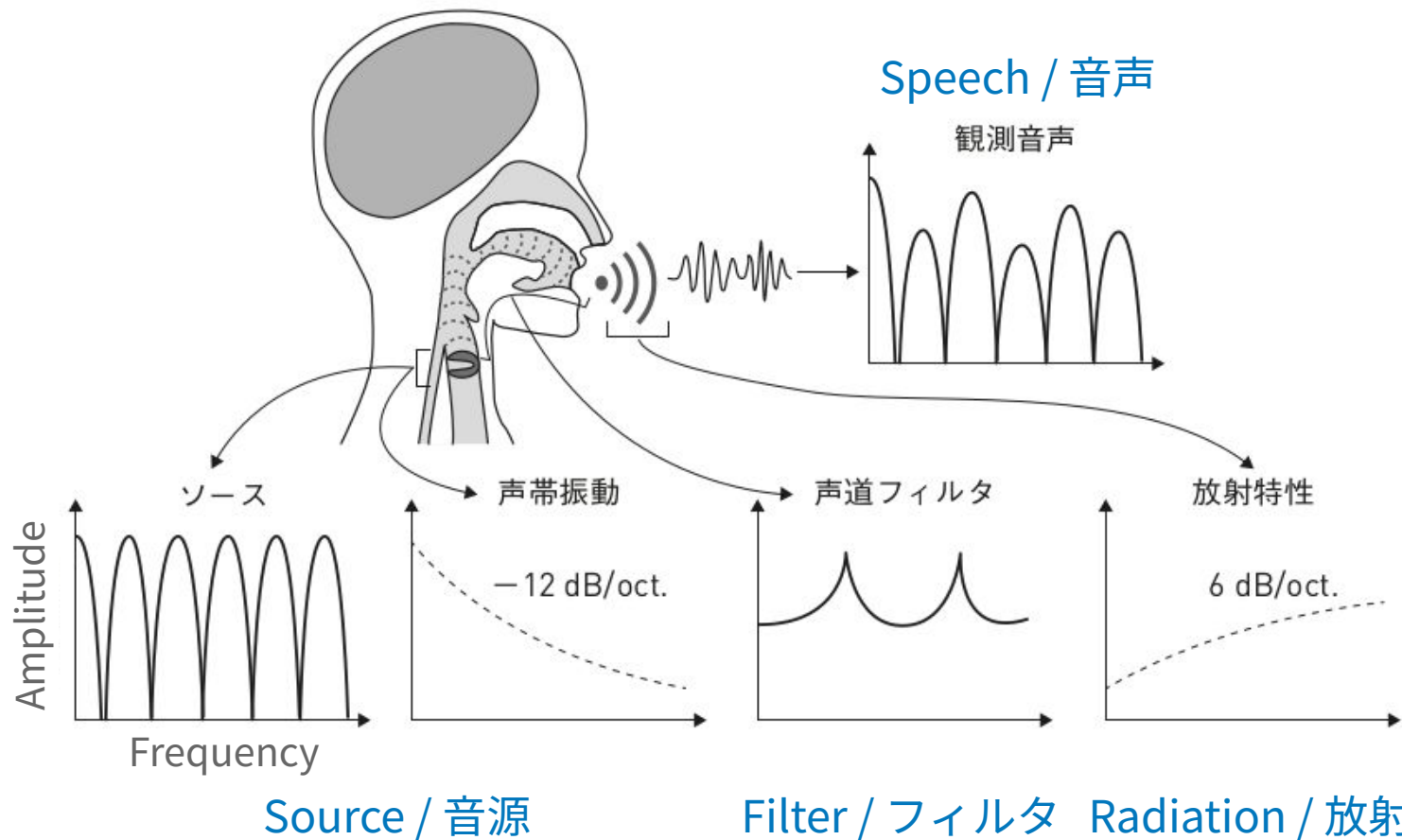
<https://www.amazon.co.jp/dp/4140113456>

- Covering 75000 Japanese words.
 - from ver.1 (1943, during WWII) to ver.5 (2016)
- Flattening of accents of loan words / 外来語のアクセント平板化
 - マイナー (minor): HLLL (1998) -> LHHH (2016)
 - フレーズ (phrase): HLLL (1998) -> LHHH (2016)
- Differences among ages / 年齢による違い
 - 甘い (*amai*, sweet): LHH (45+ yo) vs. LHL (<35 yo)

Vocoder / ボコーダ

(brief explanation)

(revisited) Source-filter model / ソース・フィルタモデル



Vocoder / ボコーダ

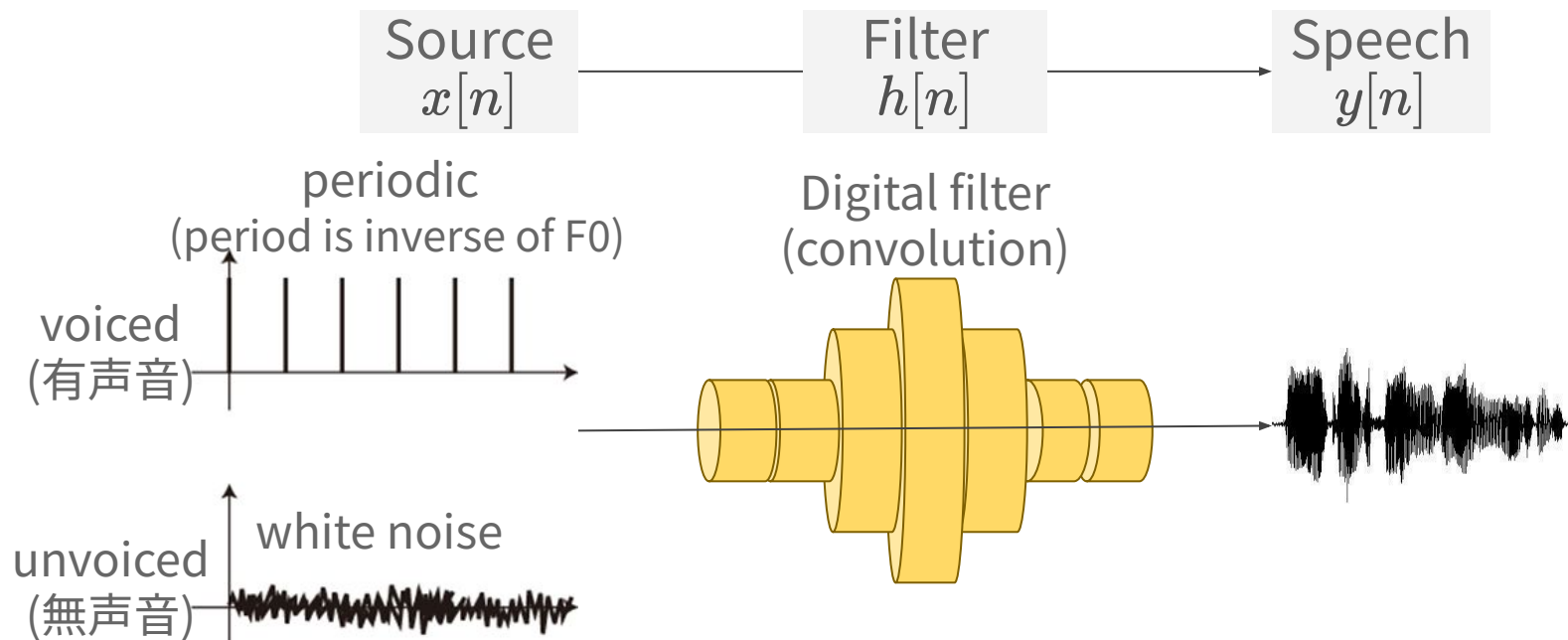
- Vocoder = voice + coder
 - Encoder: parameterize features from speech waveform
 - Decoder: synthesize speech waveform from features



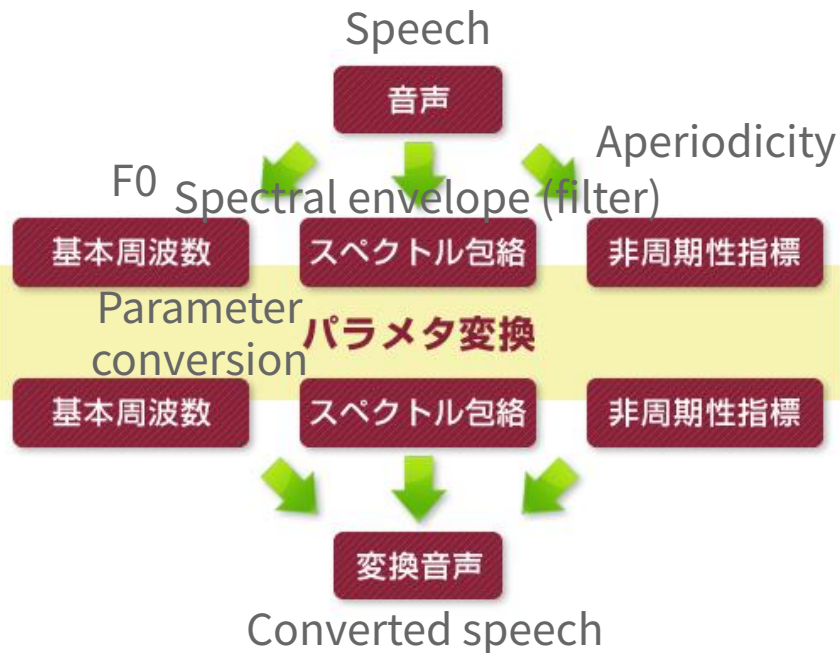
- Types
 - Phase vocoder (フェーズボコーダ): based on amplitude and instantaneous phase
 - Sinusoidal vocoder (正弦波ボコーダ): additive sine wavforms
 - Channel vocoder (チャネルボコーダ): based on source-filter model
 - Neural vocoder (ニューラルボコーダ): based on neural networks

Channel vocoder / チャネルボコーダ

Based on the source-filter theory, source and filter components are parameterized.



WORLD vocoder



<https://www.isc.meiji.ac.jp/~mmorise/world/introductions.html>

The screenshot shows a Jupyter Notebook titled "WORLD vocoder.ipynb". The notebook contains the following code cells:

```
[1] # Install Python modules
#pip install numpy scipy
!pip install librosa
!pip install pyworld # WORLD vocoder
!apt-get install sox # downsampling

[2] # Import modules
import numpy as np
from scipy.io import wavfile
import librosa as rs # Fourier analysis
import pyworld as pw # vocoder analysis
import matplotlib.pyplot as plt
from google.colab import files
import IPython as ip

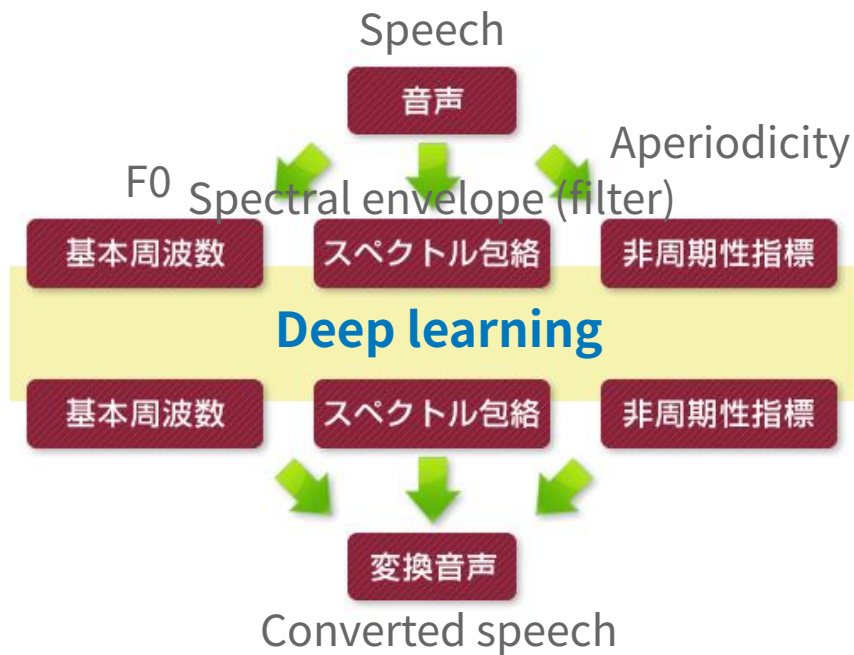
[3] # upload file
!rm -f sample.wav sample_normed.wav
uploaded = files.upload() # file upload
!sox sample.wav -c 1 -r 16000 sample_normed.wav # downsampling

[4] # load file
fs, x = wavfile.read("sample_normed.wav") # fs: sampling rate, x: waveform
x = np.asarray(x, dtype=np.float64)
ip.display.display(ip.display.Audio(x, rate = fs, autoplay=False))
```

The notebook also shows a section titled "Spectrogram".

https://colab.research.google.com/drive/1FS2Wa28ISuXxY_cvFvcHuuiSFv9v0CiH?usp=sharing (please copy to your gdrive.)

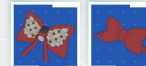
(off-topic) deep learning-based real-time speaker conversion



My system (2018)



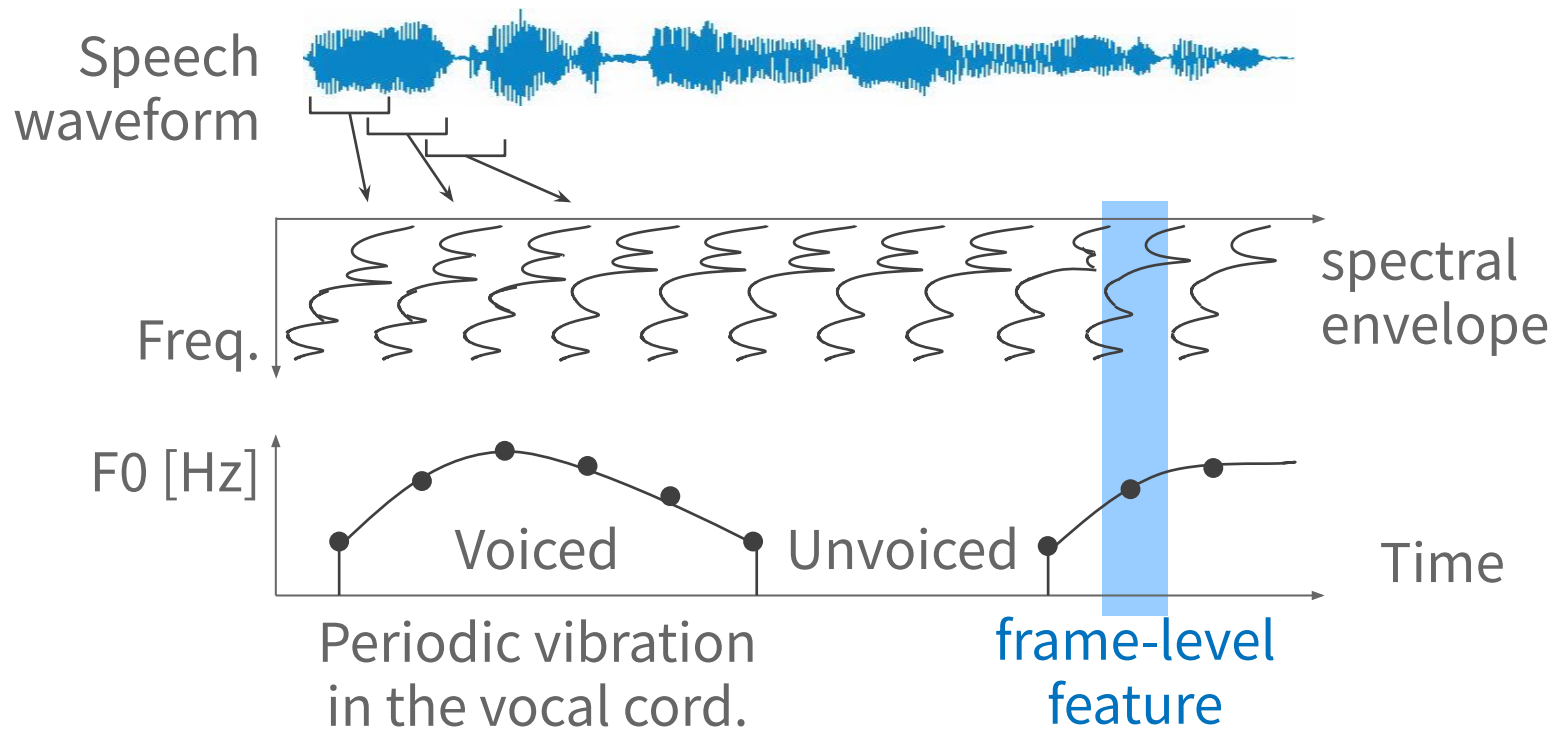
蝶ネクタイ型変声機



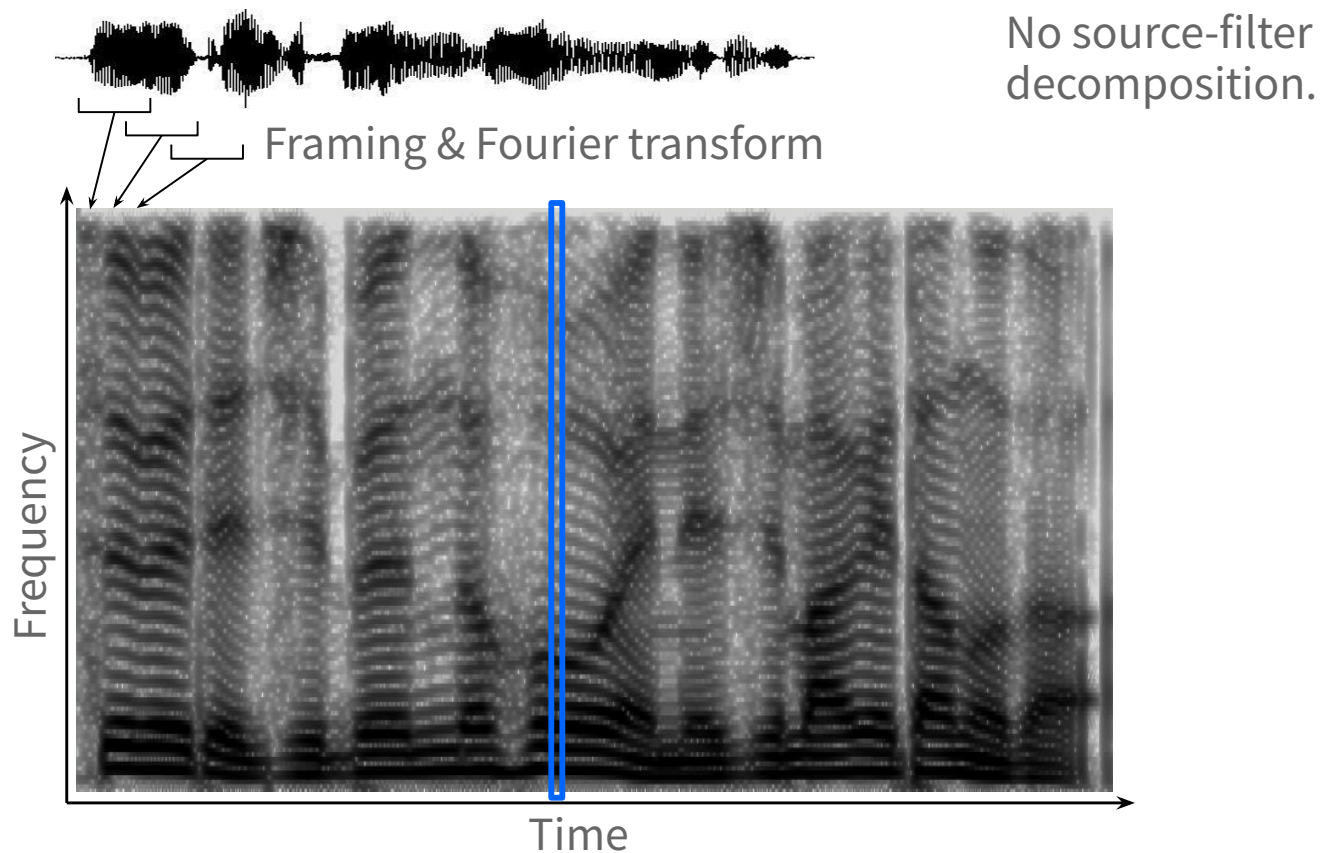
これは、ネクタイの裏についているダイヤルを調整して、発音自在に声色を変えられることのできるアイテム。右側のダイヤルで音程、左側のダイヤルで音量を調整する。サイズは小豆の大きさで調整可能。



Summary: vocoder analysis



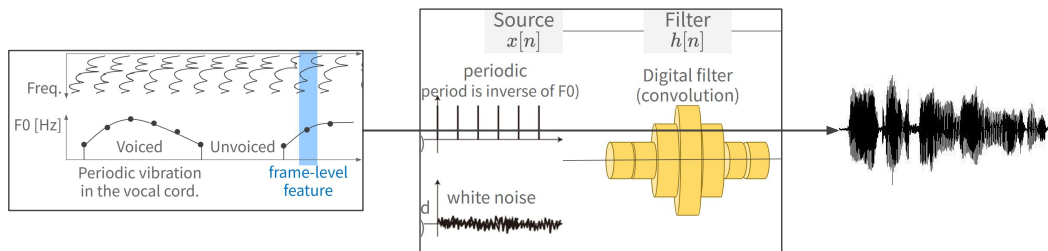
Alternative parameterization: (mel-)spectrogram



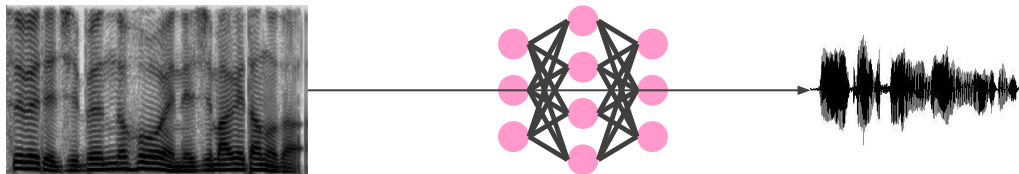
Alternative decoder: neural vocoder (HiFi-GAN, BigVGAN, etc.)

- Channel vocoder (signal processing-based)
 - light computation but has modeling error.

I will explain in the future class (possibly...)

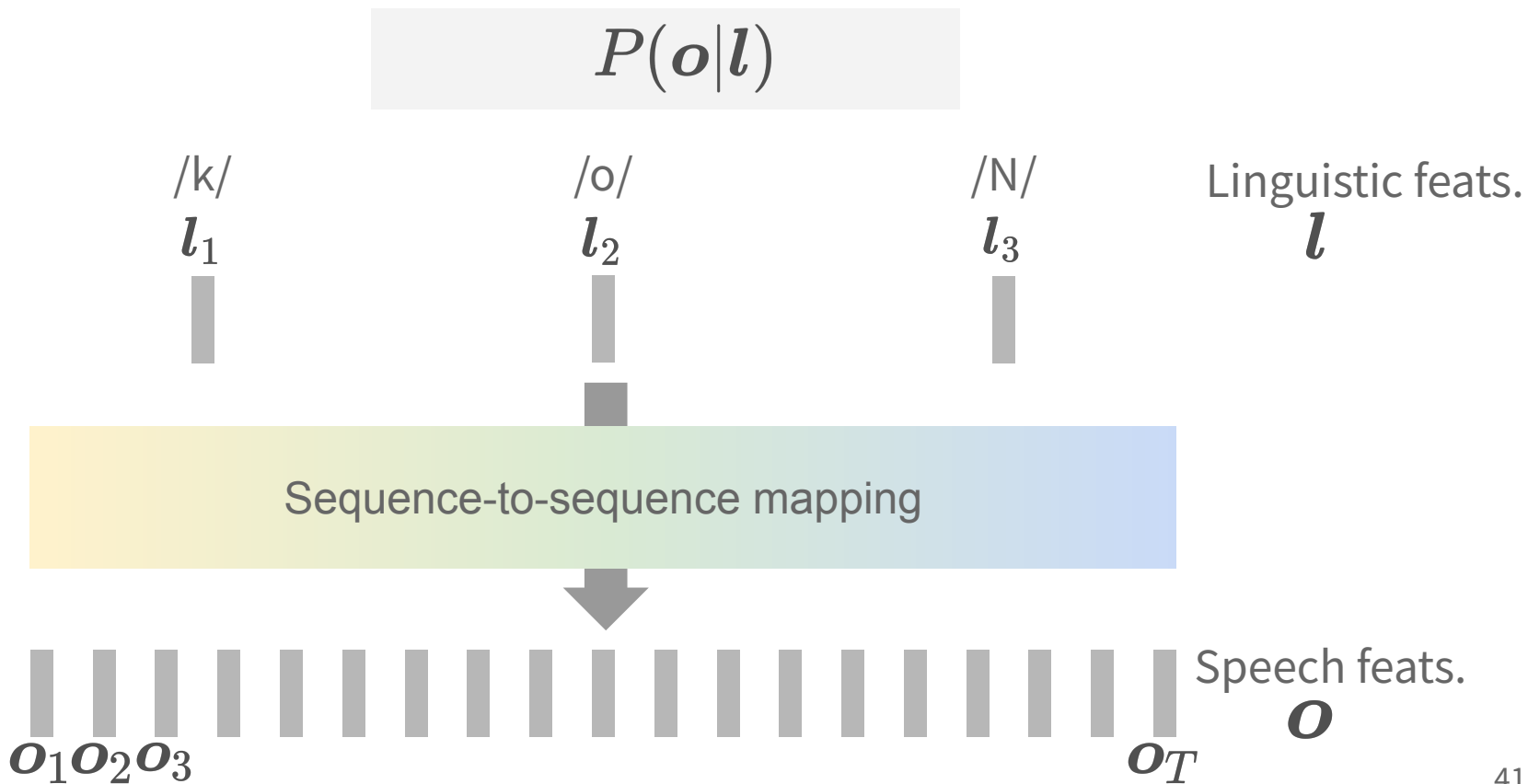


- Neural vocoder (deep learning-based, current standard)
 - heavy computation but very high quality, jointly trained with other DNNs.



Acoustic model / 音響モデル

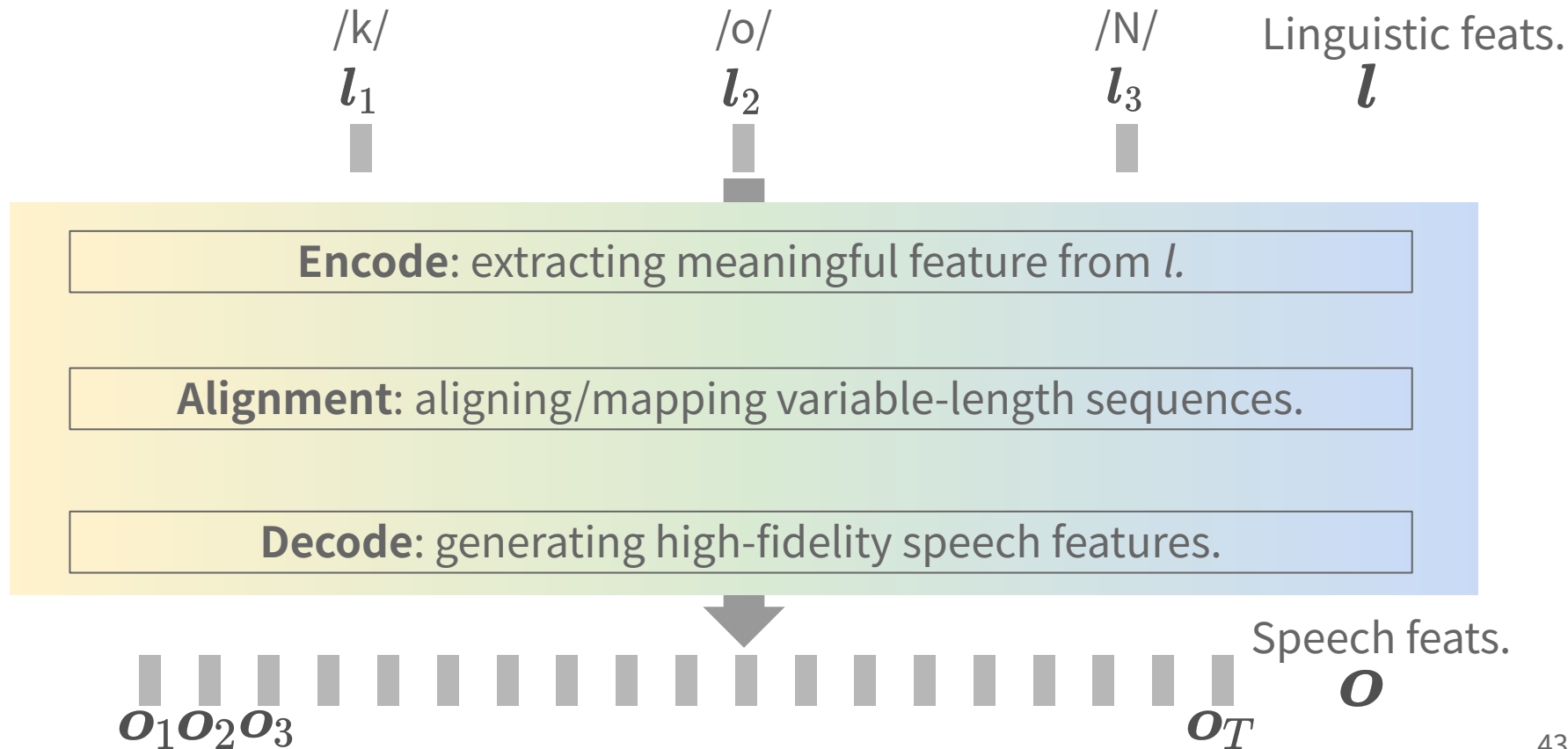
Task definition



Task definition

- length of ***l*** << length of ***O***
 - こんにちは (*konnichiwa*) -> **9** phonemes /k o N n i c h i w a/
 -  -> **200** frames (5 msec frame shift, 1 sec duration)
- Feature type:
 - Linguistic feats: **discrete** / 離散
 - Speech feats: **continuous** / 連続

Into details: three inner steps

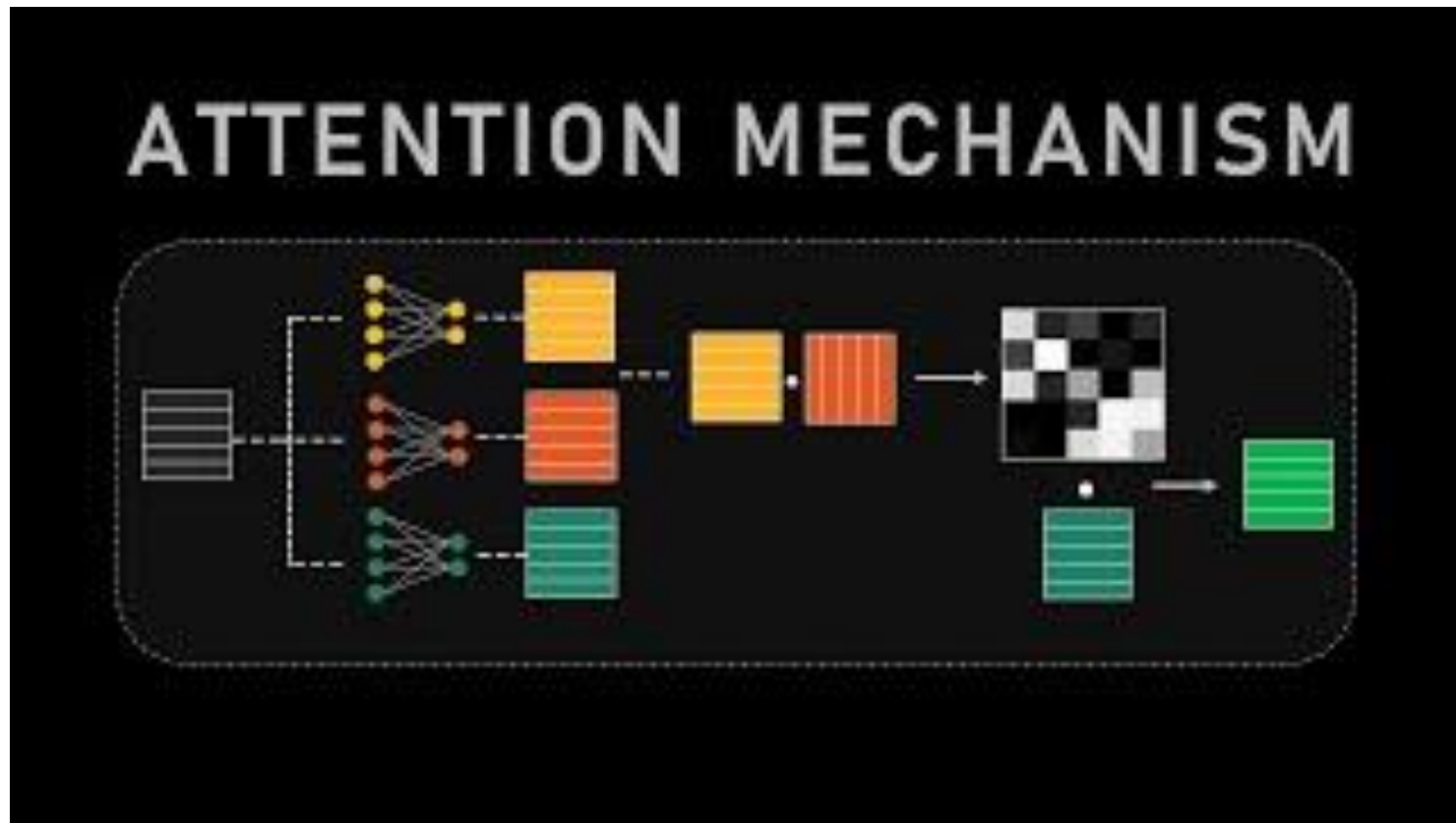


Key module: word embedding layer / 単語埋め込み層



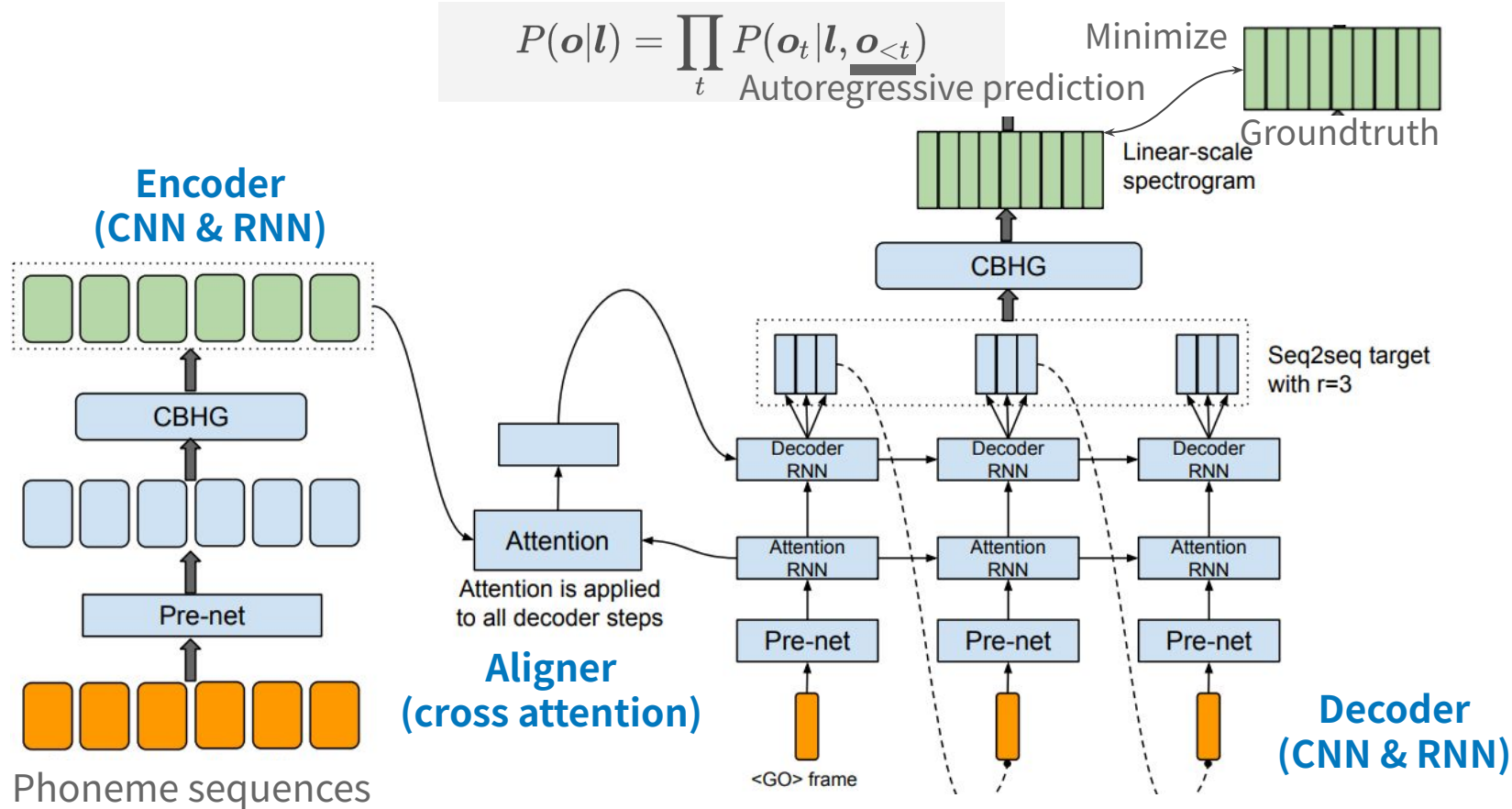
(Fundamentals of deep learning is out of scope of this class. Please study in advance.)⁴⁴

Key module: self-attention and Transformer

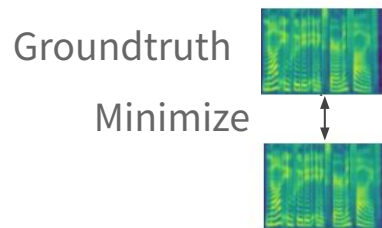


(Fundamentals of deep learning is out of scope of this class. Please study in advance.)⁴⁵

Tacotron: pioneer of end-to-end TTS

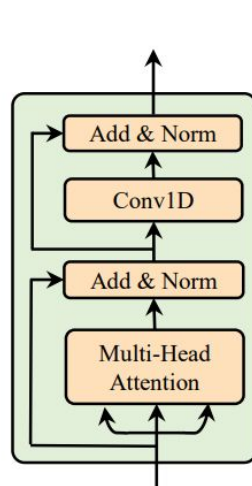
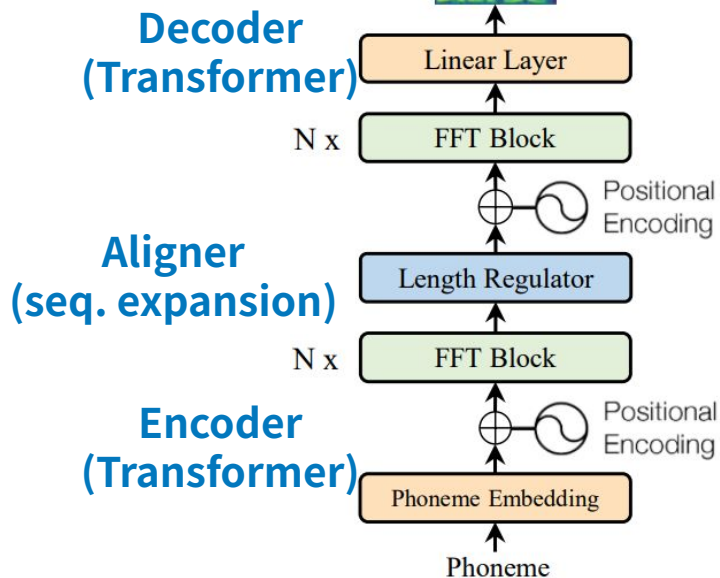


FastSpeech: duration-informed TTS

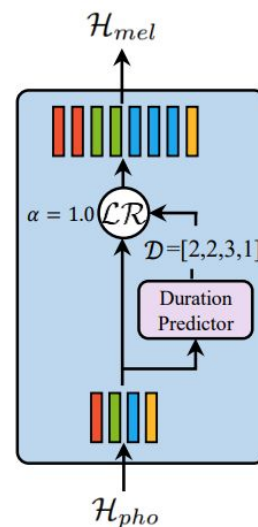


$$P(\mathbf{o}|\mathbf{l}) = \prod_t \underbrace{P(\mathbf{o}_t|\mathbf{q}, \mathbf{l})}_{\text{Feature prediction}} \underbrace{P(\mathbf{q}|\mathbf{l})}_{\text{Alignment}}$$

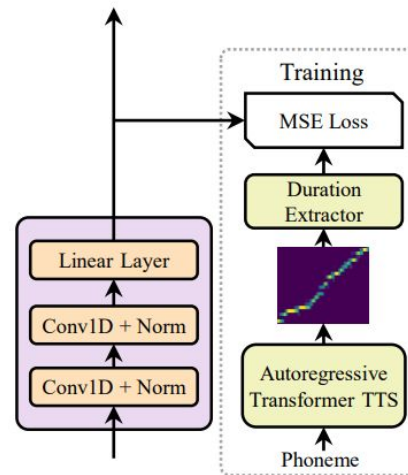
Feature prediction
under given alignment



Transformer



Sequence expansion



(a) Feed-Forward Transformer

(b) FFT Block

(c) Length Regulator

(d) Duration Predictor

Advanced topics (only abstract in today's class)

- Diffusion model / 拡散モデル
 - “Grad-TTS: A Diffusion Probabilistic Model for Text-to-Speech” (ICML2021)
- Joint training with neural vocoder / ボコーダとの同時学習
 - “JETS: Jointly Training FastSpeech2 and HiFi-GAN for End to End Text to Speech” (Interspeech2022)
- Variational autoencoder / 変分オートエンコーダ
 - “Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech” (ICML2021)
- Generative adversarial network / 敵対的生成ネットワーク

Tools

Tools

- Opensource
 - ESPnet <https://github.com/espnet/espnet>
 - Speechbrain <https://github.com/speechbrain/speechbrain>
- Products
 - ElevenLabs <https://elevenlabs.io/>
 - VoiceVox <https://voicevox.hiroshiba.jp/> (Japanese lang. only)
- Dataset (corpus)
 - LJspeech (En), JSUT (Ja), and so on.

Conclusion / まとめ

Conclusion

- History
 - From electric circuits to deep learning
- Formulation
 - Speech synthesis is essentially treated as a probabilistic task.
- Subproblems
 - Text analysis: text -> textual feature
 - Acoustic model: textual feature -> speech feature
 - Vocoder: speech feature -> speech waveform